Inspiring Innovation and Leadership

KARATINA UNIVERSITY

UNIVERSITY EXAMINATIONS
2024/2025 ACADEMIC YEAR

THIRD YEAR **SECOND** SEMESTER EXAMINATION

FOR THE DEGREE OF

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY/ BACHELOR OF SCIENCE IN COMPUTER SCIENCE

COURSE CODE: COM 311

COURSE TITLE: COMPUTER GRAPHICS

DATE: 23RD APRIL 2025 TIME: 12PM-2PM

INSTRUCTION TO CANDIDATES

- SEE INSIDE

INSTRUCTIONS:

Answer Question 1 and any other 2 questions

Show workings for all computations

QUESTION ONE (30 MARKS)

a. Describe the terms computer Graphics [2 marks]
b. Differentiate between Pixels and Texels as used in computer graphics. [2 marks]
c. Outline any FOUR sectors where computer Graphics can be applied in the industry. [4 marks]
d. Describe any FOUR types of Computer graphics [4 marks]
e. Explain the role of linear algebra in computer graphics. [4 marks]
f. List any FOUR features of OpenGL [4 marks]
g. Explain how LCD screen work in the context of computer graphics. [4 marks]
h. State the cause of antialiasing in Computer Graphics
i. Differentiate between phong and Goround shading [4 marks]
j. Explain the concepts for clipping in computer graphics. [2 marks]

QUESTION TWO (20 MARKS)

a. Differentiate between Orthographic projections are parallel projections as used in computer graphics. [4 marks]
b. Explain THREE advantages of WebGL compared to plain OpenGL. [6 marks]
c. Given that a line is expressed by the equitation y=mx+c, assume two points a(x0,y0)and b(x1,y1). Derive the equation for y and m and write an algorithm for a computer programme that will plot a line. [10 marks]

QUESTION THREE (13 MARKS)

a. Explain the following terms as used in computer graphics
    i. Graphics Pipeline. [2 marks]
    ii. Ray tracing [2 marks]
b. What is role of Normal Vectors in lighting calculations. [4 marks]

c. Differentiate between 2D and 3D graphics. [4 marks]

d. An image has the following coordinates:

$$A = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}$$

Required:

e. Compute the new coordinates of the points in the image if the image is zoomed out by a factor of 2. [4 marks]

f. Compute the new coordinates of the points if the image is reflected along the y-axis. [4 marks]

QUESTION FOUR (20 MARKS)

a. Explain TWO advantages of using a dedicated graphics processor. [4 Marks]

b. Differentiate between the following terms as used in computer graphics:

i. Synthetic and physical images as used in computer graphics. [2 marks]

ii. GLU and GLUT components of an OpenGL Library [2 marks]

iii. interlaced and non-interlaced displays [2 marks]

c. A point C on a sphere is placed such that the light bouncing on it is reflected at 90 degrees. Given that A and B are two vectors parallel to the surface of the sphere at point C, compute the surface normal at point C. [10 marks]

QUESTION FIVE (20 MARKS)

a. Write a HTML code that creates a 400px by 600px canvas. [4 marks]

b. Write WebGL code that contextualizes the canvas in (c) above [6 marks]

c. Write code that displays a rectangle filled with the colour red. [8 marks]

d. Explain how the refresh rate affect the visualization of graphic on a screens. [2 marks]